

RIX2024 - Batch 1 - Docker Practical Assessment
12576 Nisanth Saravanan

Co-Living Expense Manager

W3H Analysis

Project Title: Co-Living Expense Manager	
What?	How?
1. What are the necessary modules? A: Modules Needed: 1) Login and Registration. 2) User Dashboard.	Login and Registration Module: 1) Register User: Method 1: Use Simple Form Page. Method 2: Use a Modular Form Page. Method 3: Use Google OAuth and other 3rd party services.
2. What are the necessary functionalities of Login and Registration Modules? A: Login and Registration Module: 1) Register User. 2) Login as User.	2) Login as User: Method 1: Use Simple Form. Method 2: Use a Modular Form. Method 3: Use 3rd party services.
3. What are the necessary functionalities of User Dashboard? A: User Dashboard: 1) Perform CRUD Operation with Expenses. 2) Filter Expenses.	User Dashboard: 1) Perform CRUD Operation with Expenses: Method 1: Use Simple Forms and Buttons to perform CRUD. Method 2: Have separate pages for each operation Method 3: Use Modals.
4. What are the tables needed? A: Tables Required: • Expenses	2) Filter Expenses: Method 1: Use a simple search bar.

- Users

5. What are the fields of user table?

A: Fields of Users table:

- Id (Primary Key) - int
- Name – string
- Contact – int
- Password – string

6. What are the fields of the Expenses table?

A: Fields of Expenses Table:

- Id (Primary Key) - int
- Name – string
- Type – string
- Description - string
- Amount – float/decimal

7. What are the must have dependencies?

A: Dependencies:

- axios
- react-router-dom

8. What are the validations in the application?

A: Overall validations list:

- Frontend simple JS Validations.
- Server Side validations.

Method 2: Have tables and sorting buttons.

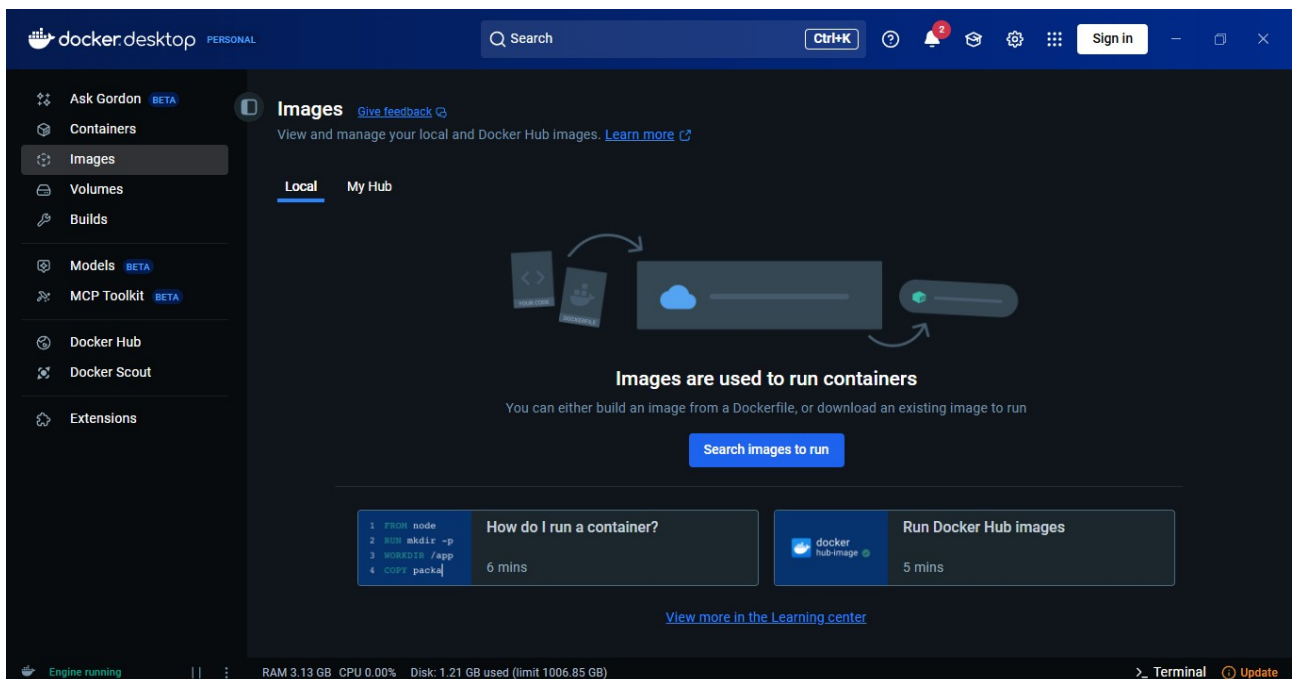
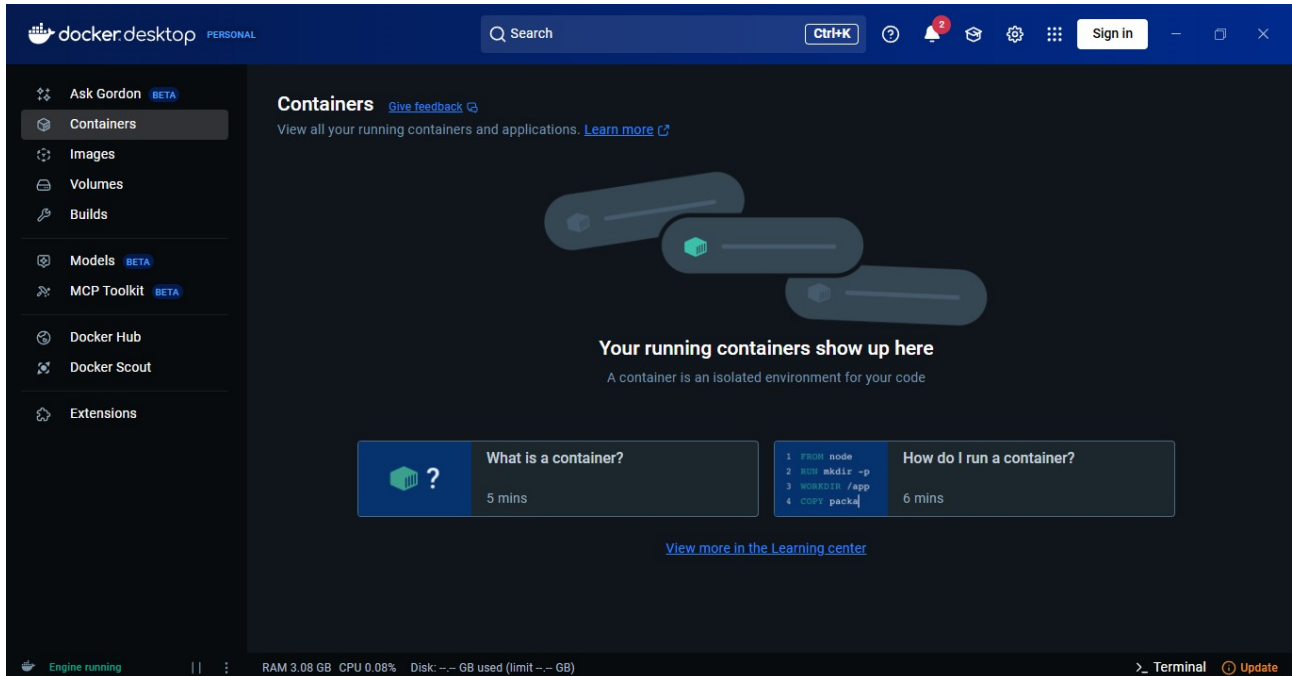
Method 3: Have a separate page to pick out exact results.

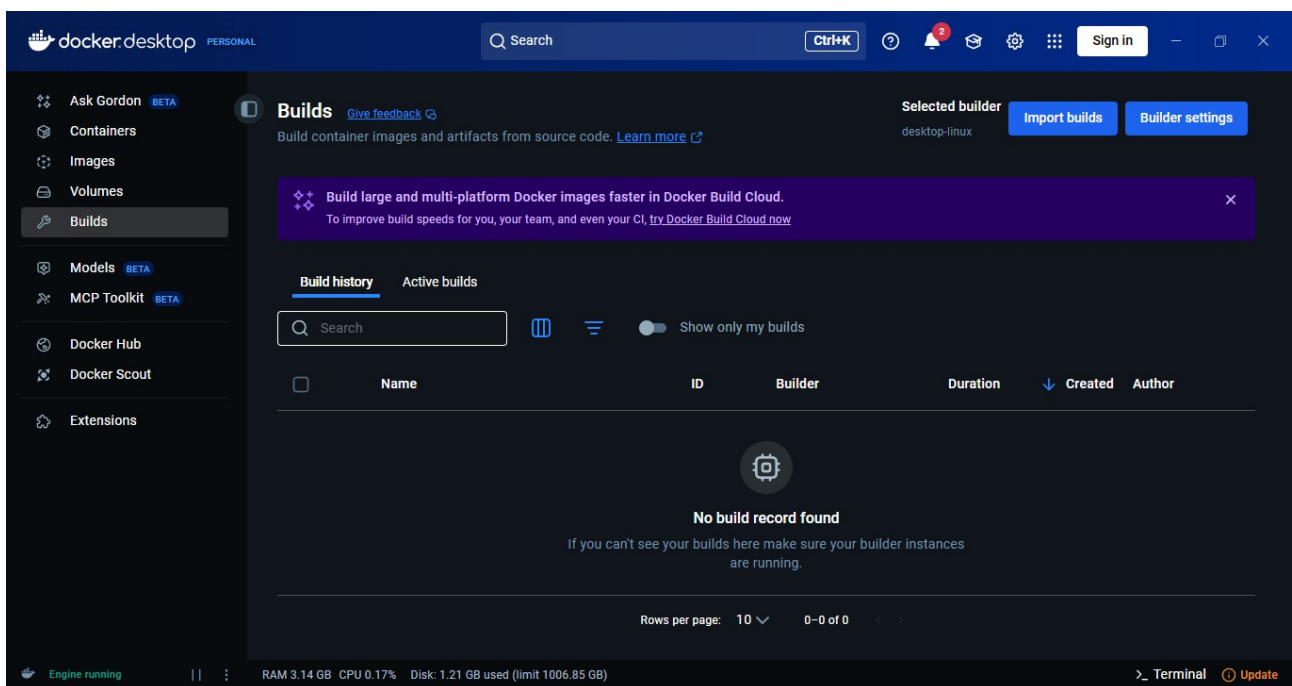
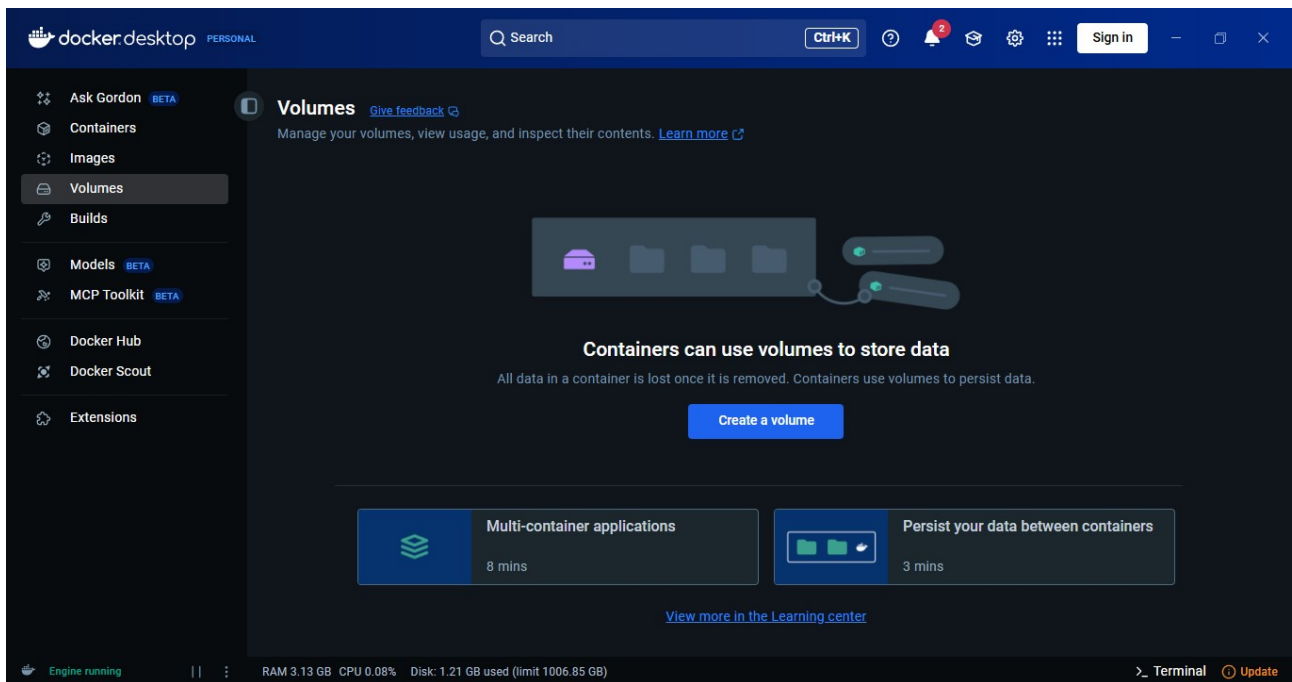
• Form Validations.	
Login and Registration Module: 1) Register User: Method 1: Use Simple Form Page. A: Since the overall scale of the application is small and data needed is rather minimal, it's wise to stick with traditional forms. 2) Login as User: Method 1: Use Simple Form. A: Since the overall scale of the application is small and data needed is rather minimal, it's wise to stick with traditional forms. User Dashboard: 1) Perform CRUD Operation with Expenses: Method 1: Use Simple Forms and Buttons to perform CRUD. A: Since the overall scale of the application is small and data needed is rather minimal, it's wise to stick with traditional forms. 2) Filter Expenses: Method 1: Use a simple search bar. A: Effective as it may, the simple input field can be directly mapped with the Expenses that are being	Login and Registration Module: 1) Register User: Method 2: Use a Modular Form Page. A: Increases the complexity of the app, and also the fact that the number of needed inputs in rather minimal. Method 3: Use Google OAuth and other 3rd party services. A: Valid method but due to the scale of the application, this method is seemingly overkill. 2) Login as User: Method 2: Use a Modular Form. A: Increases the complexity of the app, and also the fact that the number of needed inputs in rather minimal. Method 3: Use 3rd party services. A: Valid method but due to the scale of the application, this method is seemingly overkill. User Dashboard: 1) Perform CRUD Operation with Expenses: Method 2: Have separate pages for

displayed.	each operation. A: Since the admin needs all information in one place and also that the number of operations they can do is limited, it is best to have a single page control centre type. 2) Filter Expenses: Method 2: Have tables and sorting buttons. A: This method is also viable but not selected due to the slight increase in the number of code needed for its execution. Method 3: Have a separate page to pick out exact results. A: Having the all the data displayed in one page and search functionality in another is not ideal. Hence, this method is rejected.
Why?	Why Not?

Output Screenshots

Docker Desktop – Initial





Running the Docker-Compose

```
PS D:\Repos\RelevantzRIX2024\Intern Files\Intern Assessment\L1 Assessment\12576_Nisanth_Saravanan_Docker\ExpenseTrackerApplication> docker-compose up
--build
time="2025-08-19T16:10:31+05:30" level=warning msg="D:\\Repos\\RelevantzRIX2024\\Intern Files\\Intern Assessment\\L1 Assessment\\12576_Nisanth_Saravanan_Docker\\ExpenseTrackerApplication\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 8/12
 - mysql-db [██████████] 84.94MB / 235.1MB Pulling 10.3s
   ✓ ac1d09b99a58 Download complete 2.2s
   - 227c621e92bf Downloading [=====] 22.02MB/49.82MB 5.5s
   - c17e5ae8443e Downloading [=====] 34.6MB/128.9MB 5.5s
   ✓ 29def2d20dd4 Download complete 1.6s
   ✓ 21c02f1ef6a5 Download complete 1.3s
   ✓ eb4c833f6ac8 Download complete 1.7s
   ✓ d90a2df7273a Download complete 3.2s
   ✓ 6ae00a88a906 Download complete 1.6s
   - 10aec8a104c7 Downloading [=====] 22.02MB/49.5MB 5.5s
   ✓ 13ecfab825f5 Download complete 1.6s
   ✓ 0dd51a1cf370 Download complete 1.5s
```

```
#9 [aspnet-app runtime 1/3] FROM mcr.microsoft.com/dotnet/aspnet:8.0@sha256:6d40ae71165cb8a4427477ae3b447b05c080330cabae08b0dae72552bcd34d8
#9 sha256:15113326d88088b488ac1ff098f080de963f9ec99ae9e9b8aaa893bb9fbb3 0B / 11.09MB 0.2s
#9 sha256:2b6f82574228686d9b3e8e387f1802ad71840af1f7e060f5ac630142f1cc5fb5 0B / 154B 0.2s
#9 sha256:8824f6aeff3745bb49d3e7b7a03c642826c1f43a89329c02a8ae0688867a78bf 0B / 32.25MB 0.2s
#9 sha256:d58a7b951b0567343615d6d5cb94598356fe8bd9d8c49d27b3914fe6187c4b66 0B / 3.28kB 0.2s
#9 sha256:15113326d88088b488ac1ff098f080de963f9ec99ae9e9b8aaa893bb9fbb3 2.10MB / 11.09MB 0.5s
#9 sha256:2b6f82574228686d9b3e8e387f1802ad71840af1f7e060f5ac630142f1cc5fb5 154B / 154B 0.3s done
#9 sha256:d58a7b951b0567343615d6d5cb94598356fe8bd9d8c49d27b3914fe6187c4b66 3.28kB / 3.28kB 0.3s done
#9 sha256:16bf5aa6abccdc9ead9f7ec5c05492f7bffd1800933d3971902baf3c27d270 0B / 18.74MB 0.2s
#9 sha256:b1badc6e50664185acfaa0ca25d8076061c2a9d881cecaaad281ae11af000ce 0B / 28.23MB 0.2s
#9 sha256:15113326d88088b488ac1ff098f080de963f9ec99ae9e9b8aaa893bb9fbb3 3.15MB / 11.09MB 0.8s
#9 sha256:16bf5aa6abccdc9ead9f7ec5c05492f7bffd1800933d3971902baf3c27d270 1.05MB / 18.74MB 0.5s
#9 sha256:15113326d88088b488ac1ff098f080de963f9ec99ae9e9b8aaa893bb9fbb3 4.19MB / 11.09MB 1.1s
#9 sha256:8824f6aeff3745bb49d3e7b7a03c642826c1f43a89329c02a8ae0688867a78bf 2.10MB / 32.25MB 0.9s
#9 sha256:16bf5aa6abccdc9ead9f7ec5c05492f7bffd1800933d3971902baf3c27d270 2.10MB / 18.74MB 0.6s
#9 sha256:15113326d88088b488ac1ff098f080de963f9ec99ae9e9b8aaa893bb9fbb3 5.24MB / 11.09MB 1.2s
#9 sha256:b1badc6e50664185acfaa0ca25d8076061c2a9d881cecaaad281ae11af000ce 2.10MB / 28.23MB 0.8s
#9 sha256:16bf5aa6abccdc9ead9f7ec5c05492f7bffd1800933d3971902baf3c27d270 3.15MB / 18.74MB 0.9s
#9 sha256:15113326d88088b488ac1ff098f080de963f9ec99ae9e9b8aaa893bb9fbb3 6.29MB / 11.09MB 1.4s
#9 sha256:8824f6aeff3745bb49d3e7b7a03c642826c1f43a89329c02a8ae0688867a78bf 4.19MB / 32.25MB 1.4s
#9 sha256:16bf5aa6abccdc9ead9f7ec5c05492f7bffd1800933d3971902baf3c27d270 4.19MB / 18.74MB 1.2s
#9 sha256:b1badc6e50664185acfaa0ca25d8076061c2a9d881cecaaad281ae11af000ce 4.19MB / 28.23MB 1.2s
#9 sha256:15113326d88088b488ac1ff098f080de963f9ec99ae9e9b8aaa893bb9fbb3 7.34MB / 11.09MB 1.7s
#9 sha256:16bf5aa6abccdc9ead9f7ec5c05492f7bffd1800933d3971902baf3c27d270 5.24MB / 18.74MB 1.4s
#9 sha256:15113326d88088b488ac1ff098f080de963f9ec99ae9e9b8aaa893bb9fbb3 8.39MB / 11.09MB 1.8s
#9 sha256:8824f6aeff3745bb49d3e7b7a03c642826c1f43a89329c02a8ae0688867a78bf 6.29MB / 32.25MB 1.8s
```

```
#13 [frontend internal] load build context
#13 transferring context: 56.70MB 44.2s
#13 ...

#18 [aspnet-app build 2/6] WORKDIR /src
#18 DONE 1.6s

#13 [frontend internal] load build context
#13 ...

#19 [aspnet-app build 3/6] COPY ExpenseTrackerWebApi.csproj ./
#19 DONE 0.1s

#13 [frontend internal] load build context
```

```

#28 [frontend build 5/6] COPY . .
#28 DONE 7.2s

#29 [frontend build 6/6] RUN npm run build
#29 0.844
#29 0.844 > expense-management-frontend@0.1.0 build
#29 0.844 > react-scripts build
#29 0.844
#29 2.483 Creating an optimized production build...

```

Success

```

#32 [frontend] resolving provenance for metadata file
#32 DONE 0.0s
[+] Running 8/8
  ✓ expensetrackerapplication-frontend      Built      0.0s
  ✓ expensetrackerapplication-aspnet-app    Built      0.0s
  ✓ Network expensetrackerapplication_testnetwork Created    0.1s
  ✓ Network expensetrackerapplication_default Created    0.1s
  ✓ Volume "expensetrackerapplication_mysqlvolume" Created    0.0s
  ✓ Container mysql-container              Created    0.2s
  ✓ Container aspnet-container              Created    0.2s
  ✓ Container react_frontend               Created    0.2s
Attaching to aspnet-container, mysql-container, react_frontend
mysql-container | 2025-08-19 10:44:10+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.43-1.el9 started.
mysql-container | 2025-08-19 10:44:11+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
mysql-container | 2025-08-19 10:44:11+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.43-1.el9 started.
mysql-container | 2025-08-19 10:44:11+00:00 [Note] [Entrypoint]: Initializing database files
mysql-container | 2025-08-19T10:44:11.526965Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a
future release. Please use SET GLOBAL host_cache_size=0 instead.
mysql-container | 2025-08-19T10:44:11.527215Z 0 [System] [MY-013169] [Server] /usr/sbin/mysqld (mysqld 8.0.43) initializing of server in progress as
process 80
mysql-container | 2025-08-19T10:44:11.546326Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
react_frontend | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration

```

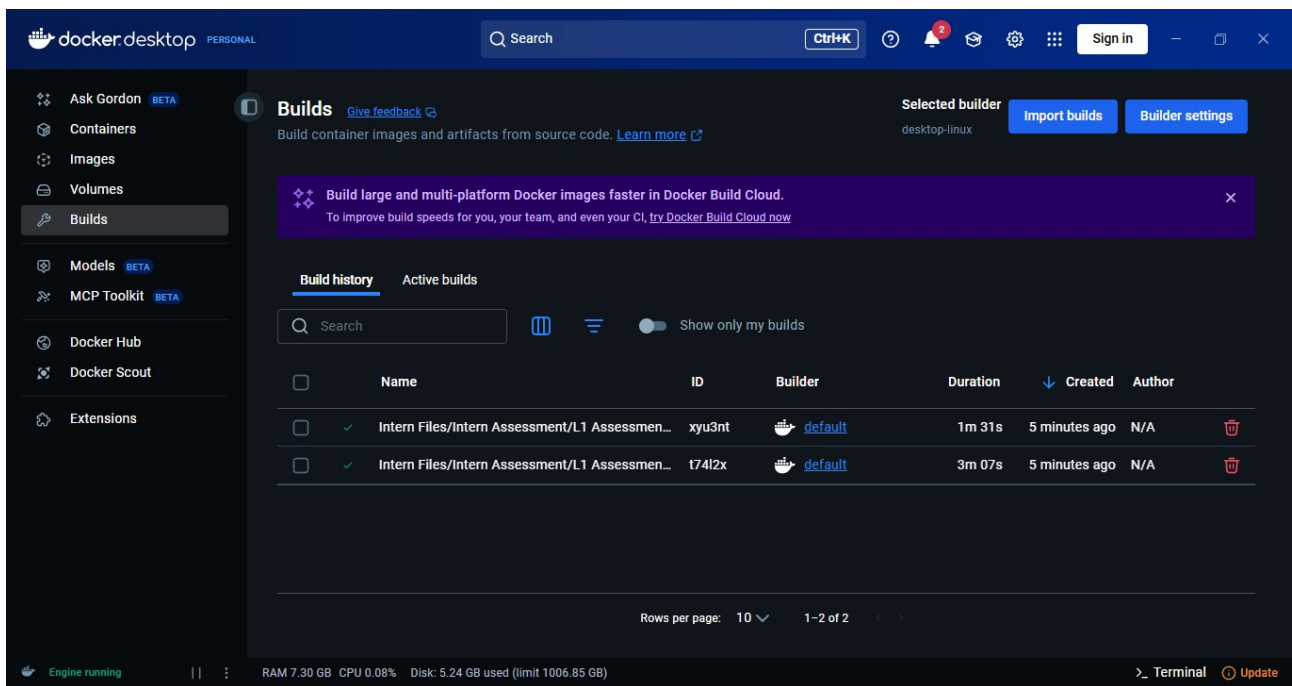
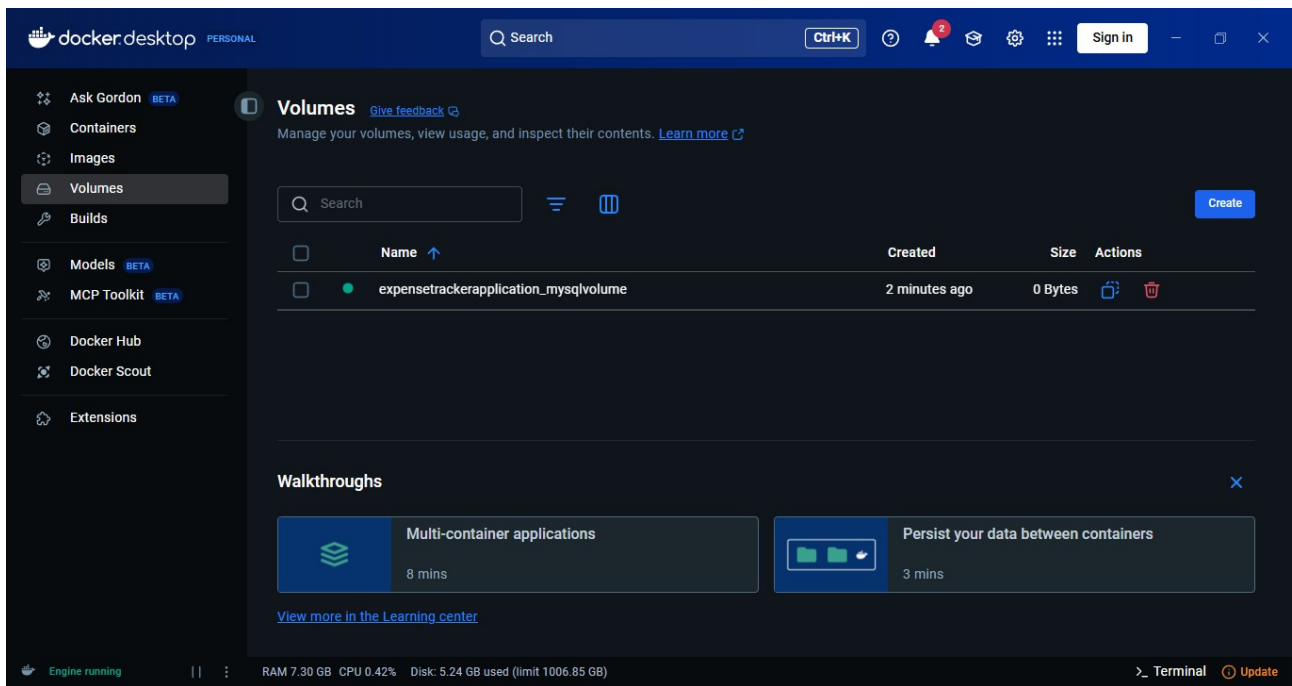
Docker Desktop After a Successful Run

The screenshot shows the Docker Desktop interface with the 'Containers' tab selected. The left sidebar contains navigation options: Ask Gordon, Containers, Images, Volumes, Builds, Models, MCP Toolkit, Docker Hub, Docker Scout, and Extensions. The main panel displays 'Containers' with a search bar and a toggle for 'Only show running containers'. It shows four running containers: expensetracker, mysql-contair, aspnet-contair, and react_frontend. Above the table, system resource usage is shown: Container CPU usage at 0.71% / 1200% and Container memory usage at 444.29MB / 7.4GB. The bottom status bar indicates 'Engine running' and system resources: RAM 7.27 GB, CPU 0.00%, and Disk 5.24 GB used.

	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	expensetracker	-	-	-	0.61%	39 seconds ago	[Stop] [Refresh] [Delete]
☐	mysql-contair	28712be154ed	mysql:8.0	3307:3306	0.6%	1 minute ago	[Stop] [Refresh] [Delete]
☐	aspnet-contair	c25cb8579280	expensetracker	5001:5000	0.01%	39 seconds ago	[Stop] [Refresh] [Delete]
☐	react_frontend	cb315033132c	expensetracker	3000:80	0%	1 minute ago	[Stop] [Refresh] [Delete]

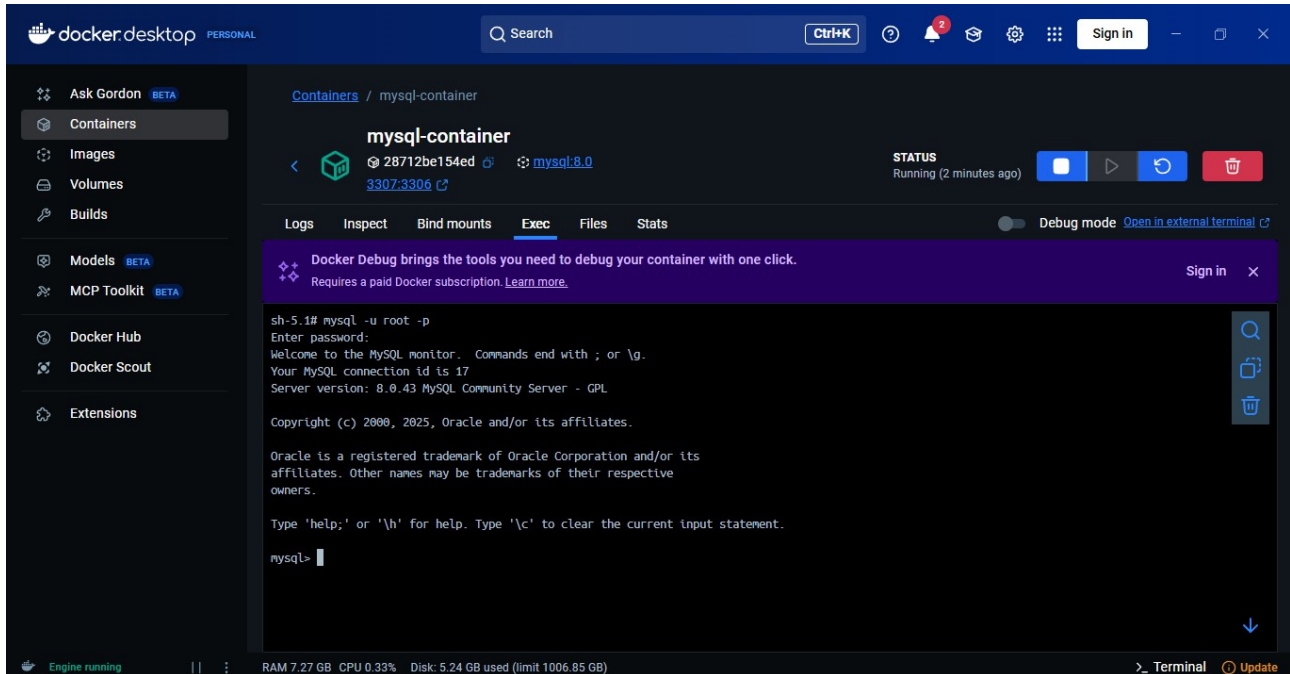
The screenshot shows the Docker Desktop interface with the 'Images' tab selected. The left sidebar is the same as the previous screenshot. The main panel displays 'Images' with tabs for 'Local' and 'My Hub'. It shows three local images: mysql, expensetrackerapplication-as, and expensetrackerapplication-frt. Above the table, it shows '1.12 GB / 2.83 GB in use' and '3 images'. A 'Delete' button and 'Space to be reclaimed 831.61 MB' are visible. The bottom status bar shows 'Engine running' and system resources: RAM 7.28 GB, CPU 0.08%, and Disk 5.24 GB used.

	Name	Tag	Image ID	Created	Size	Actions
☒	mysql	8.0	f483084b3708	28 days ago	1.06 GB	[Play] [Refresh] [Delete]
☐	expensetrackerapplication-as	latest	0bf840e27f69	3 minutes ago	332.45 MB	[Play] [Refresh] [Delete]
☐	expensetrackerapplication-frt	latest	fbda2387a886	1 minute ago	81.66 MB	[Play] [Refresh] [Delete]



Containerized Outputs

Database

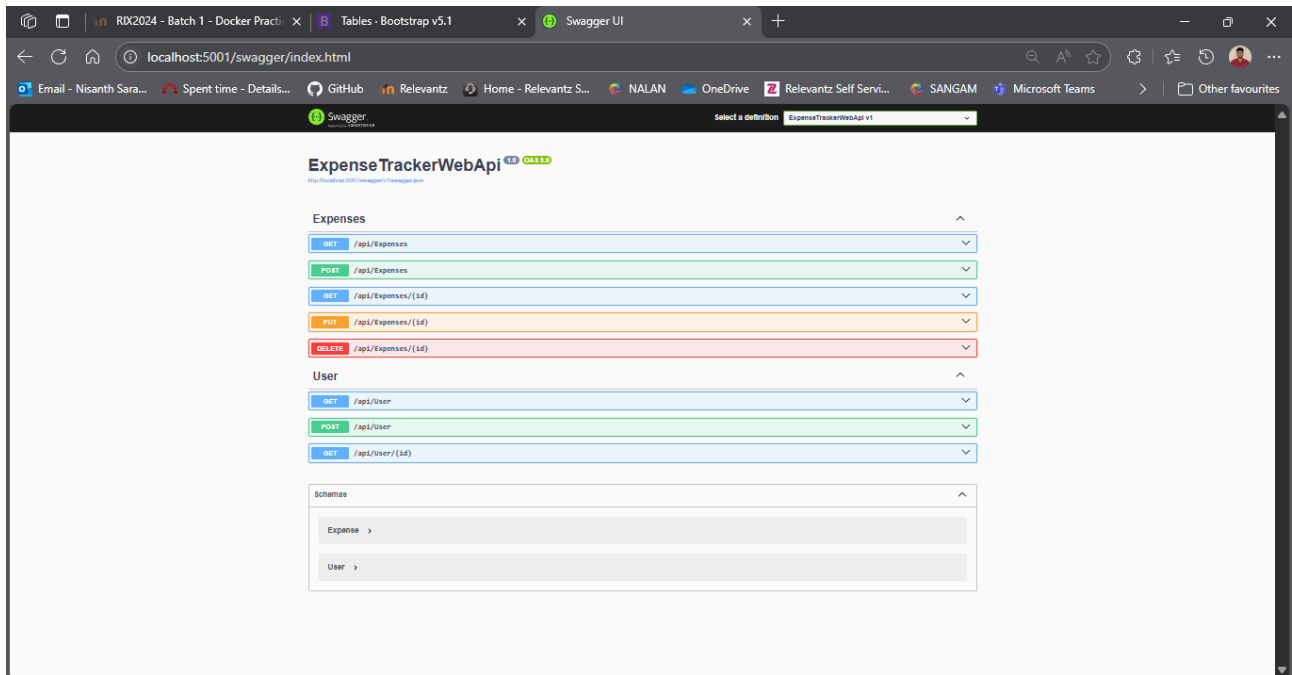


```
mysql> show databases;
+-----+
| Database |
+-----+
| ExpenseDb |
| information_schema |
| mysql |
| performance_schema |
| productdb |
| sys |
+-----+
6 rows in set (0.01 sec)
```

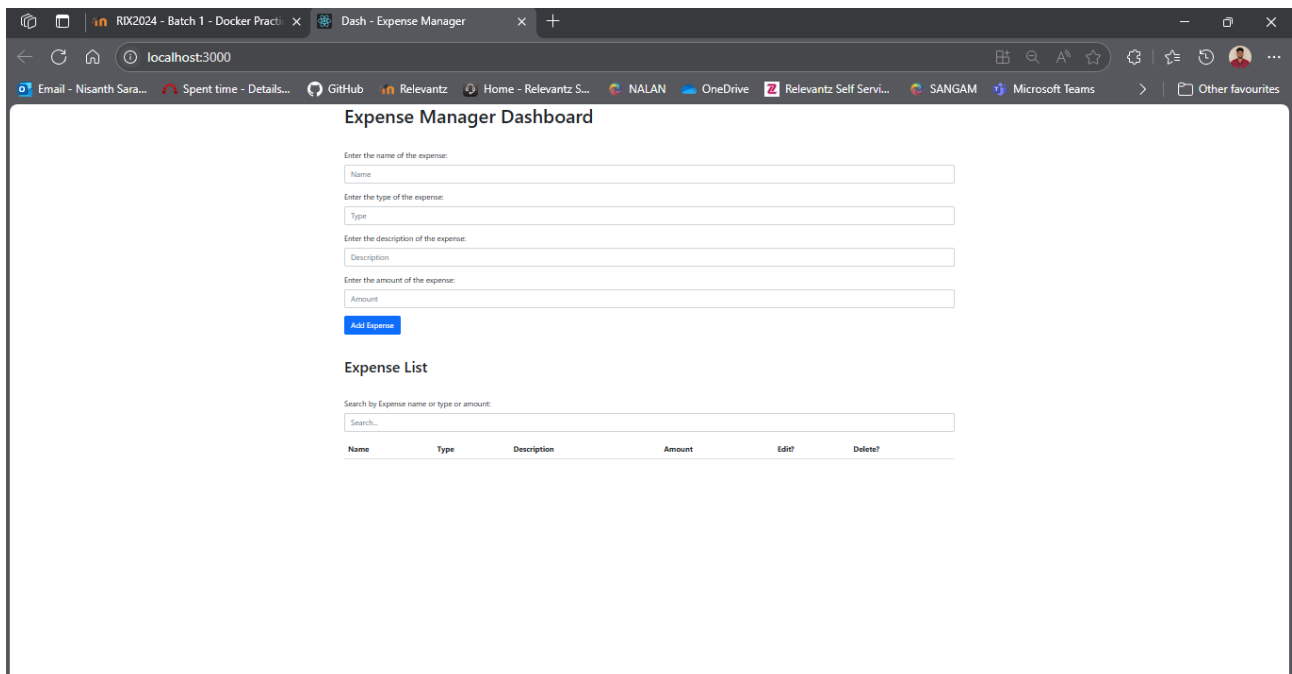
```
mysql> use ExpenseDb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> show tables;
+-----+
| Tables_in_ExpenseDb |
+-----+
| Expenses |
| Users |
| __EFMigrationsHistory |
+-----+
3 rows in set (0.00 sec)
```

Web API



React Frontend



CRUD Operations

Create - Without Description

Expense Manager Dashboard

Enter the name of the expense:

Enter the type of the expense:

Enter the description of the expense:

Enter the amount of the expense:

Expense List

Search by Expense name or type or amount:

Name	Type	Description	Amount	Edit?	Delete?
------	------	-------------	--------	-------	---------

```
mysql> use ExpenseDb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> select * from Expenses;
+----+-----+-----+-----+-----+
| Id | Name          | Type    | Description | Price |
+----+-----+-----+-----+-----+
| 2  | Electricity Bills | Necessity | No Description. | 250000 |
+----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Create – With Description

Expense Manager Dashboard

Enter the name of the expense:

Enter the type of the expense:

Enter the description of the expense:

Enter the amount of the expense:

Add Expense

Expense List

Search by Expense name or type or amount:

Name	Type	Description	Amount	Edit?	Delete?
Electricity Bills	Necessity	No Description.	Rs.250000	<button>Edit</button>	<button>Delete</button>

```
mysql> select * from Expenses;
```

```
+-----+-----+-----+-----+
| Id | Name           | Type    | Description                | Price |
+-----+-----+-----+-----+
| 2  | Electricity Bills | Necessity | No Description.            | 250000 |
| 3  | Veggies         | Food    | Tomatoes still left on the list | 134 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Edit

Expense List

Search by Expense name or type or amount:

Search...

Name	Type	Description	Amount	Edit?	Delete?
Electricity Bills	Necessity	No Description.	Rs.250000	Edit	Delete
Veggies	Food	Tomatoes still left on the list	Rs.134	Edit	Delete

From

Expense Manager Dashboard

Enter the name of the expense:

Electricity Bills

Enter the type of the expense:

Necessity

Enter the description of the expense:

No Description.

Enter the amount of the expense:

250000

Update Expense

Cancel

To

Expense Manager Dashboard

Enter the name of the expense:

Electricity Bills

Enter the type of the expense:

Necessity

Enter the description of the expense:

The next month bill is 10% off.

Enter the amount of the expense:

250000

Update Expense

Cancel

Expense List

Search by Expense name or type or amount:

Search...					
Name	Type	Description	Amount	Edit?	Delete?
Electricity Bills	Necessity	The next month bill is 10% off.	Rs.250000	<button>Edit</button>	<button>Delete</button>
Veggies	Food	Tomatoes still left on the lst	Rs.134	<button>Edit</button>	<button>Delete</button>

From

```
mysql> select * from Expenses;
+----+-----+-----+-----+-----+
| Id | Name          | Type   | Description          | Price |
+----+-----+-----+-----+-----+
| 2  | Electricity Bills | Necessity | No Description.      | 250000 |
| 3  | Veggies         | Food    | Tomatoes still left on the lst | 134 |
+----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

To

```
mysql> select * from Expenses;
+----+-----+-----+-----+-----+
| Id | Name          | Type   | Description          | Price |
+----+-----+-----+-----+-----+
| 2  | Electricity Bills | Necessity | The next month bill is 10% off. | 250000 |
| 3  | Veggies         | Food    | Tomatoes still left on the lst | 134 |
+----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

Delete

Before

```
mysql> select * from Expenses;
+----+-----+-----+-----+-----+
| Id | Name          | Type   | Description                                     | Price |
+----+-----+-----+-----+-----+
| 2  | Electricity Bills | Necessity | The next month bill is 10% off. | 250000 |
| 3  | Veggies          | Food    | Tomatoes still left on the 1st | 134   |
+----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

Expense List

Search by Expense name or type or amount:

Search...

Name	Type	Description	Amount	Edit?	Delete?
Electricity Bills	Necessity	The next month bill is 10% off.	Rs.250000	<button>Edit</button>	<button>Delete</button>
Veggies	Food	Tomatoes still left on the 1st	Rs.134	<button>Edit</button>	<button>Delete</button>

After

```
mysql> select * from Expenses;
+----+-----+-----+-----+-----+
| Id | Name  | Type | Description                                     | Price |
+----+-----+-----+-----+-----+
| 3  | Veggies | Food | Tomatoes still left on the 1st | 134   |
+----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Expense List

Search by Expense name or type or amount:

Search...

Name	Type	Description	Amount	Edit?	Delete?
Veggies	Food	Tomatoes still left on the 1st	Rs.134	<button>Edit</button>	<button>Delete</button>

Searching

Sample Data

Expense List

Search by Expense name or type or amount:

Name	Type	Description	Amount	Edit?	Delete?
Veggies	Food	Tomatoes still left on the list	Rs.134	Edit	Delete
Electricity Bills	Necessity	No Description.	Rs.250000	Edit	Delete
Water Bills	Necessity	No Description.	Rs.250000	Edit	Delete
School Fee	Education	No Description.	Rs.250000	Edit	Delete

Searching By Name

Expense List

Search by Expense name or type or description:

Name	Type	Description	Amount	Edit?	Delete?
Electricity Bills	Necessity	No Description.	Rs.250000	Edit	Delete
Water Bills	Necessity	No Description.	Rs.250000	Edit	Delete

Searching By Type

Expense List

Search by Expense name or type or description:

Name	Type	Description	Amount	Edit?	Delete?
Veggies	Food	Tomatoes still left on the list	Rs.134	Edit	Delete

Expense List

Search by Expense name or type or description:

Name	Type	Description	Amount	Edit?	Delete?
School Fee	Education	No Description.	Rs.250000	Edit	Delete

Searching By Description

Expense List

Search by Expense name or type or description:

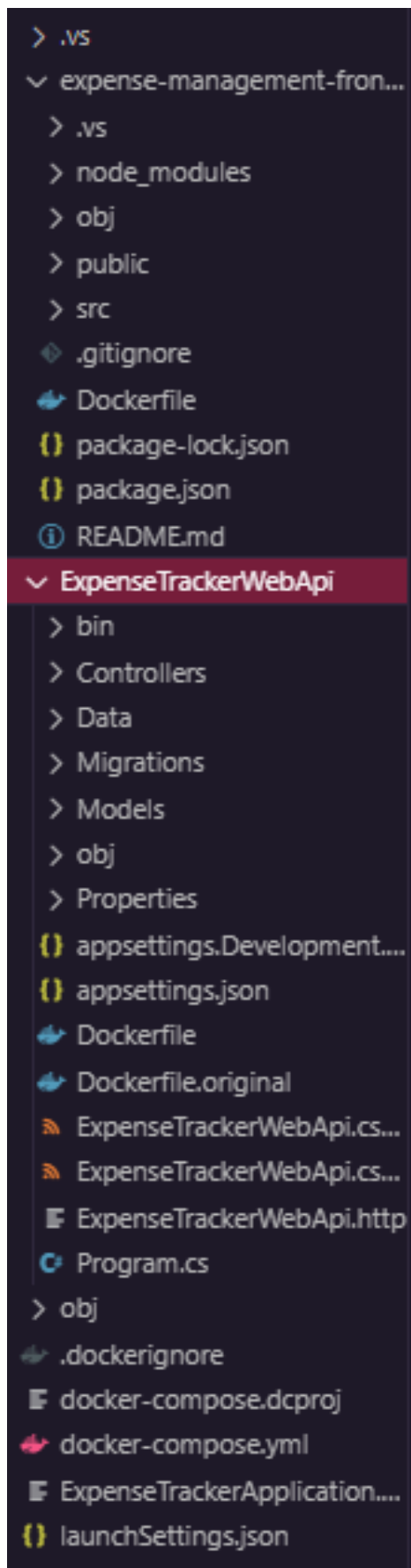
Name	Type	Description	Amount	Edit?	Delete?
Electricity Bills	Necessity	No Description.	Rs.250000	Edit	Delete
Water Bills	Necessity	No Description.	Rs.250000	Edit	Delete
School Fee	Education	No Description.	Rs.250000	Edit	Delete

Expense List

Search by Expense name or type or description:

Name	Type	Description	Amount	Edit?	Delete?
Veggies	Food	Tomatoes still left on the list	Rs.134	Edit	Delete

Codebase



docker-compose.yml

```
version: "3.9"
services:

  aspnet-app:
    build:
      context: ./ExpenseTrackerWebApi
      dockerfile: Dockerfile
    container_name: aspnet-container
    ports:
      - "5001:5000" # Host:Container port mapping
    depends_on:
      - mysql-db
    environment:
      - ASPNETCORE_ENVIRONMENT=Development
      - ASPNETCORE_URLS=http://+:5128
      - ConnectionStrings__DefaultConnection=Server=mysql-
db;Database=ExpenseDb;User=root;Password=123456;Port=3306
    networks:
      - testnetwork
    restart: on-failure

  frontend:
    build:
      context: ./expense-management-frontend
    container_name: react_frontend
    ports:
      - "3000:80"
    depends_on:
      - aspnet-app

  mysql-db:
    image: mysql:8.0
    container_name: mysql-container
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: 123456
      MYSQL_DATABASE: productdb
    ports:
      - "3307:3306"
    volumes:
      - mysqlvolume:/var/lib/mysql
    networks:
      - testnetwork

networks:
  testnetwork:
```

```
volumes:  
  mysqlvolume:
```

.dockerignore

```
*/.classpath
*/.dockerignore
*/.env
*/.git
*/.gitignore
*/.project
*/.settings
*/.toolstarget
*/.vs
*/.vscode
*//*.*proj.user
*//*.*dbmdl
*//*.*jfm
*/azds.yaml
*/bin
*/charts
*/docker-compose*
*/Dockerfile*
*/node_modules
*/npm-debug.log
*/obj
*/secrets.dev.yaml
*/values.dev.yaml
LICENSE
README.md
!*/.gitignore
!.git/HEAD
!.git/config
!.git/packed-refs
!.git/refs/heads/**
```

Back-end ASP .NET Web API

DockerFile

```
# Build stage
FROM mcr.microsoft.com/dotnet/sdk:8.0 AS build
WORKDIR /src

# Copy and restore dependencies
COPY ExpenseTrackerWebApi.csproj ./
RUN dotnet restore

# Copy the rest of the source code
COPY . ./
RUN dotnet publish -c Release -o /app/publish

# Runtime stage
FROM mcr.microsoft.com/dotnet/aspnet:8.0 AS runtime
WORKDIR /app

# Copy published output from build stage
COPY --from=build /app/publish .

# Expose port 5000 for the API
EXPOSE 5000

# Set environment variable for ASP.NET Core to listen on port 5000
ENV ASPNETCORE_URLS=http://+:5000

# Run the application
ENTRYPOINT ["dotnet", "ExpenseTrackerWebApi.dll"]
```

Models

User

```
using System.ComponentModel.DataAnnotations;

namespace ExpenseTrackerWebApi.Models
{
    public class User
    {
        [Key]
        public int Id { get; set; }

        public string? Name { get; set; }

        public int? Contact { get; set; }

        public string? Password { get; set; }
    }
}
```

Expense

```
using System.ComponentModel.DataAnnotations;

namespace ExpenseTrackerWebApi.Models
{
    public class Expense
    {
        [Key]
        public int Id { get; set; }

        public string? Name { get; set; }

        public string? Type { get; set; }

        public string? Description { get; set; }

        public float? Price { get; set; }
    }
}
```

Data

```
using Microsoft.EntityFrameworkCore;
using ExpenseTrackerWebApi.Models;

namespace ExpenseTrackerWebApi.Data
{
    public class ApplicationDbContext : DbContext
    {
        public ApplicationDbContext(DbContextOptions<ApplicationDbContext> options)
            : base(options) { }

        public DbSet<User> Users { get; set; }
        public DbSet<Expense> Expenses { get; set; }
    }
}
```

AppSettings.json

```
{
  "ConnectionStrings": {
    "DefaultConnection":
"Server=mysql;Port=3306;Database=ExpenseDb;User=root;Password=12345;"
  },
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*"
}
```

Controllers

UserController

```
using Microsoft.AspNetCore.Mvc;
using Microsoft.EntityFrameworkCore;
using ExpenseTrackerWebApi.Data;
using ExpenseTrackerWebApi.Models;

namespace ExpenseTrackerWebApi.Controllers
{
    [ApiController]
    [Route("api/[controller]")]
    public class UserController : ControllerBase
    {
        private readonly ApplicationDbContext _context;

        public UserController(ApplicationDbContext context)
        {
            _context = context;
        }

        // GET: api/users
        [HttpGet]
        public async Task<IActionResult> GetAll()
        {
            var user = await _context.Users.ToListAsync();
            return Ok(user);
        }

        // GET: api/users/{id}
        [HttpGet("{id}")]
        public async Task<IActionResult> Get(int id)
        {
            var user = await _context.Users.FindAsync(id);
            if (user == null) return NotFound();
            return Ok(user);
        }

        // POST: api/users
        [HttpPost]
        public async Task<IActionResult> Create([FromBody] User user)
        {
            if (!ModelState.IsValid) return BadRequest(ModelState);

            _context.Users.Add(user);
            await _context.SaveChangesAsync();
        }
    }
}
```

```
        return CreatedAtAction(nameof(Get), new { id = user.Id }, user);  
    }  
}
```

Expense Controller

```
using Microsoft.AspNetCore.Mvc;
using Microsoft.EntityFrameworkCore;
using ExpenseTrackerWebApi.Data;
using ExpenseTrackerWebApi.Models;

namespace ExpenseTrackerWebApi.Controllers
{
    [ApiController]
    [Route("api/[controller]")]
    public class ExpensesController : ControllerBase
    {
        private readonly ApplicationDbContext _context;

        public ExpensesController(ApplicationDbContext context)
        {
            _context = context;
        }

        // GET: api/expenses
        [HttpGet]
        public async Task<IActionResult> GetAll()
        {
            var expenses = await _context.Expenses.ToListAsync();
            return Ok(expenses);
        }

        // GET: api/expenses/{id}
        [HttpGet("{id}")]
        public async Task<IActionResult> Get(int id)
        {
            var expense = await _context.Expenses.FindAsync(id);
            if (expense == null) return NotFound();
            return Ok(expense);
        }

        // POST: api/expenses
        [HttpPost]
        public async Task<IActionResult> Create([FromBody] Expense expense)
        {
            if (!ModelState.IsValid) return BadRequest(ModelState);

            _context.Expenses.Add(expense);
            await _context.SaveChangesAsync();

            return CreatedAtAction(nameof(Get), new { id = expense.Id }, expense);
        }
    }
}
```

```

// PUT: api/expenses/{id}
[HttpPut("{id}")]
public async Task<IActionResult> Update(int id, [FromBody] Expense updated)
{
    var expense = await _context.Expenses.FindAsync(id);
    if (expense == null) return NotFound();

    expense.Name = updated.Name;
    expense.Type = updated.Type;
    expense.Description = updated.Description;
    expense.Price = updated.Price;

    await _context.SaveChangesAsync();
    return NoContent();
}

// DELETE: api/expenses/{id}
[HttpDelete("{id}")]
public async Task<IActionResult> Delete(int id)
{
    var expense = await _context.Expenses.FindAsync(id);
    if (expense == null) return NotFound();

    _context.Expenses.Remove(expense);
    await _context.SaveChangesAsync();
    return NoContent();
}
}
}
}

```

Program.cs

```
using Microsoft.EntityFrameworkCore;
using ExpenseTrackerWebApi.Data;
using ExpenseTrackerWebApi.Models;

var builder = WebApplication.CreateBuilder(args);

// Load connection string from appsettings.json
var connectionString =
builder.Configuration.GetConnectionString("DefaultConnection");

// Register DbContext with MySQL
builder.Services.AddDbContext<ApplicationDbContext>(options =>
    options.UseMySQL(connectionString, ServerVersion.AutoDetect(connectionString)));

// Enable CORS for React frontend
builder.Services.AddCors(options =>
{
    options.AddPolicy("AllowReactApp",
        policy => policy.WithOrigins("http://localhost:3000")
            .AllowAnyHeader()
            .AllowAnyMethod());
});

// Add services
builder.Services.AddControllers();
builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen();

// Force app to listen on port 5000
builder.WebHost.UseUrls("http://*:5000");

var app = builder.Build();

// Apply migrations on startup
using (var scope = app.Services.CreateScope())
{
    var db = scope.ServiceProvider.GetRequiredService<ApplicationDbContext>();
    db.Database.Migrate();
}

app.UseSwagger();
app.UseSwaggerUI();

app.UseCors("AllowReactApp");

app.UseAuthorization();
```

```
app.MapControllers();
```

```
app.Run();
```

Frontend – React App

DockerFile

```
# Dockerfile for React frontend
FROM node:18-alpine AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
RUN npm run build

# Serve the build with nginx
FROM nginx:alpine
COPY --from=build /app/build /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <link rel="icon" href="%PUBLIC_URL%/favicon.ico" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <meta name="theme-color" content="#000000" />
  <meta name="description"
    content="Web site created using create-react-app" />
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.1.3/dist/css/bootstrap.min.css"
rel="stylesheet" integrity="sha384-
1BmE4kWBq78iYhFldvKuhfTAU6auU8tT94WrHftjDbrCEXSU1oBoqyl2QvZ6jIW3"
crossorigin="anonymous">
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.1.3/dist/js/bootstrap.bundle.min.js"
integrity="sha384-ka7Sk0Gln4gmtz2MlQnikT1wXgYsOg+OMhuP+IlRH9sENBO0LRn5q+8nbTov4+1p"
crossorigin="anonymous"></script>
  <link rel="apple-touch-icon" href="%PUBLIC_URL%/logo192.png" />
  <!--
    manifest.json provides metadata used when your web app is installed on a
    user's mobile device or desktop. See
https://developers.google.com/web/fundamentals/web-app-manifest/
  -->
  <link rel="manifest" href="%PUBLIC_URL%/manifest.json" />
  <!--
    Notice the use of %PUBLIC_URL% in the tags above.
    It will be replaced with the URL of the `public` folder during the build.
    Only files inside the `public` folder can be referenced from the HTML.

    Unlike "/favicon.ico" or "favicon.ico", "%PUBLIC_URL%/favicon.ico" will
    work correctly both with client-side routing and a non-root public URL.
    Learn how to configure a non-root public URL by running `npm run build`.
  -->
  <title>Expense Tracker</title>
</head>
<body>
  <noscript>You need to enable JavaScript to run this app.</noscript>
  <div id="root"></div>
  <!--
    This HTML file is a template.
    If you open it directly in the browser, you will see an empty page.

    You can add webfonts, meta tags, or analytics to this file.
    The build step will place the bundled scripts into the <body> tag.

    To begin the development, run `npm start` or `yarn start`.
```

To create a production bundle, use `npm run build` or `yarn build`.

-->

</body>

</html>

index.js

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import './index.css';
import App from './App';
import reportWebVitals from './reportWebVitals';
import { BrowserRouter } from 'react-router-dom';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <React.StrictMode>
    <BrowserRouter>
      <App />
    </BrowserRouter>
  </React.StrictMode>
);

// If you want to start measuring performance in your app, pass a function
// to log results (for example: reportWebVitals(console.log))
// or send to an analytics endpoint. Learn more: https://bit.ly/CRA-vitals
reportWebVitals();
```

App.js

```
import './App.css';
import { Routes, Route } from 'react-router-dom';
import ExpensePage from './pages/ExpensePage';

function App() {
  return (
    <>
      <ExpensePage/>
    </>
  );
}

export default App;
```

ExpensePage.jsx

```
import React, { useState, useEffect } from "react";
import axios from "axios";

const ExpensePage = () => {
  const [expenses, setExpenses] = useState([]);
  const [search, setSearch] = useState("");
  const [editingExpense, setEditingExpense] = useState(null);

  const fetchexpenses = async () => {
    const response = await axios.get("http://localhost:5001/api/expenses");
    setExpenses(response.data);
  };

  useEffect(() => {
    document.title = "Dash - Expense Manager";

    fetchexpenses();
  }, [expenses]);

  const handleAddOrUpdateExpense = async (e) => {
    e.preventDefault();
    const formData = new FormData(e.target);
    const expenseData = Object.fromEntries(formData.entries());

    if (editingExpense) {
      const response = await axios.put(
        `http://localhost:5001/api/expenses/${editingExpense.id}`,
        expenseData
      );
      setExpenses(
        expenses.map((e) => (e.id === editingExpense.id ? response.data : e))
      );
      setEditingExpense(null);
    } else {
      if (expenseData.description == null || expenseData.description === "") {
        expenseData.description = "No Description.";
      }

      const response = await axios.post(
        "http://localhost:5001/api/expenses",
        expenseData
      );
      setExpenses([...expenses, response.data]);
    }

    e.target.reset();
  };
};
```

```

const handleEdit = (expense) => {
  setEditingExpense(expense);
};

const handleDelete = async (id) => {
  await axios.delete(`http://localhost:5001/api/expenses/${id}`);
  setExpenses(expenses.filter((e) => e.id !== id));
};

const filteredexpenses = expenses.filter(
  (e) =>
    e.name?.toLowerCase().includes(search.toLowerCase()) ||
    e.type?.toLowerCase().includes(search.toLowerCase()) ||
    e.description?.toLowerCase().includes(search.toLowerCase())
);

return (
  <>
    <div class="container wrapper">
      <h1 style={{ marginBottom: "3rem" }}>Expense Manager Dashboard</h1>
      <form
        onSubmit={handleAddOrUpdateExpense}
        style={{ marginBottom: "3rem" }}
      >
        <div class="mb-3">
          <label for="name" class="form-label">
            Enter the name of the expense:
          </label>
          <input
            name="name"
            placeholder="Name"
            defaultValue={editingExpense?.name || ""}
            required
            class="form-control"
          />
        </div>
        <div class="mb-3">
          <label for="type" class="form-label">
            Enter the type of the expense:
          </label>
          <input
            name="type"
            placeholder="Type"
            defaultValue={editingExpense?.type || ""}
            required
            class="form-control"
          />
        </div>
        <div class="mb-3">
          <label for="description" class="form-label">

```

```

    Enter the description of the expense:
  </label>
  <input
    name="description"
    placeholder="Description"
    defaultValue={editingExpense?.description || ""}
    class="form-control"
  />
</div>
<div class="mb-3">
  <label for="price" class="form-label">
    Enter the amount of the expense:
  </label>
  <input
    name="price"
    placeholder="Amount"
    defaultValue={editingExpense?.price || ""}
    required
    class="form-control"
  />
</div>
<button type="submit" class="btn btn-primary">
  {editingExpense ? "Update" : "Add"} Expense
</button>
{editingExpense && (
  <button
    type="button"
    class="btn btn-danger"
    onClick={() => setEditingExpense(null)}
  >
    Cancel
  </button>
)}
</form>

```

```

{/* Expense List */}
<h2 style={{ marginBottom: "3rem" }}>Expense List</h2>
{/* Search */}
<div class="mb-3" style={{ marginBottom: "3rem" }}>
  <label for="search" class="form-label">
    Search by Expense name or type or description:
  </label>
  <input
    type="text"
    placeholder="Search..."
    value={search}
    onChange={(e) => setSearch(e.target.value)}
    class="form-control"
  />
</div>
<table class="table table-hover">

```

```

    <thead>
      <tr>
        <th scope="col">Name</th>
        <th scope="col">Type</th>
        <th scope="col">Description</th>
        <th scope="col">Amount</th>
        <th scope="col">Edit?</th>
        <th scope="col">Delete?</th>
      </tr>
    </thead>
    <tbody>
      {filteredexpenses.map((e) => (
        <tr key={e.id}>
          <td>{e.name}</td>
          <td>{e.type}</td>
          <td>{e.description}</td>
          <td>Rs.{e.price}</td>
          <td>
            <button onClick={() => handleEdit(e)} class="btn btn-primary">
              Edit
            </button>
          </td>
          <td>
            <button
              onClick={() => handleDelete(e.id)}
              class="btn btn-danger"
            >
              Delete
            </button>
          </td>
        </tr>
      )})}
    </tbody>
  </table>
</div>
</>
);
};

export default ExpensePage;

```

launcSettings.json

```
{
  "profiles": {
    "Docker Compose": {
      "commandName": "DockerCompose",
      "commandVersion": "1.0",
      "serviceActions": {
        "expensetrackerwebapi": "StartDebugging"
      }
    }
  }
}
```